

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : {FACULTYNAME}

Semester : SEMESTER - 6

Subject Code & Name: BTECPC601 - INTERNET OF THINGS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

Which of the following is *not* a characteristic of IoT?

- A) Scalability
- B) Interoperability
- C) Offline Operation
- D) Heterogeneity

REST API is based on which protocol?

- A) FTP
- B) HTTP
- C) SMTP
- D) TCP/IP only

3 Which technology helps IoT handle and analyze large amounts of data?

- A) Bluetooth
- B) Cloud computing
- C) Big data analytics
- D) Embedded systems

4 Which data type in C is used to store a character?

- A) int
- B) char
- C) float
- D) string

5 What is the function of a “for” loop in C?

- A) Making decisions
- B) Calling functions
- C) Repeating tasks
- D) Declaring variables

6 The NodeMCU board is based on which chip?

- A) ATmega328
- B) ESP32
- C) ESP8266
- D) STM32

7 **NodeMCU supports programming in which IDE with Arduino core?**

- A) PyCharm
- B) Eclipse
- C) Arduino IDE
- D) Keil

8 **Which data type is used in Python to store a group of items in key-value pairs?**

- A) List
- B) Tuple
- C) Set
- D) Dictionary

9 **Which command is used in Linux to check internet connectivity?**

- A) cd
- B) dir
- C) ping
- D) copy

10 **MQTT stands for:**

- A) Message Query Transfer Technology
- B) Message Queue Telemetry Transport
- C) Multi Query Transfer Tool
- D) Multiple Quality Transfer Technology

11 **Which protocol is specifically designed for low-energy devices like wearables?**

- A) Ethernet
- B) BLE
- C) UART
- D) AMQP

12 **What is the role of GPIO pins in Raspberry Pi?**

- A) Power management
- B) RAM control
- C) Input/Output operations
- D) Booting the OS

Q2. Solve the following.

- A) Explain the characteristics of IoT?
- B) Compare Python Lists and Dictionaries with examples?

Q3. Solve the following.

- A) Explain the steps to interface NodeMCU with a web service?
- B) Compare MQTT and HTTP in context of IoT communication?

Q4. Solve Any Two of the following.

- A) **Explain the logical design of IoT?**
- B) **Explain enabling technologies for IoT?**

- C) **Write a C program to find the sum of the first 10 natural numbers using a loop?** 6

Q5. Solve Any Two of the following.

- A) **Write a Python program to find the largest of three numbers?** 6
B) **Explain GPIO and how to interface sensors with Raspberry Pi?** 6
C) **Explain how to publish data to a web server using NodeMCU?** 6

Q6. Solve Any Two of the following.

- A) **Explain MQTT protocol with diagram and use case?** 6
B) **Compare BLE, UART, and Ethernet in IoT applications?** 6
C) **Explain steps to configure Wi-Fi in Raspberry Pi?** 6